



Drug-induced QT Prolongation Atlas (DIQTA) for enhancing cardiotoxicity management

Shihai Li^a, Zili Xu^a, Mingkun Guo^a, Menglong Li^a, Zhining Wen^{a,b,*}

^a College of Chemistry, Sichuan University, Chengdu, Sichuan 610064, China

^b Medical Big Data Center, Sichuan University, Chengdu, Sichuan 610064, China

Drug-induced prolongation of the QT interval is common in a variety of pharmaceutical treatments and can lead to serious clinical outcomes. Although substantial efforts have been made to prevent drug-induced QT interval prolongation, the lack of a centralized data source remains the main obstacle to further study of the underlying mechanism and the development of effective prediction strategies. To fill this gap, we propose a schema for stratifying the risk of marketed QT prolonging drugs based on US Food and Drug Administration (FDA)-approved drug labeling and developed a Drug-Induced QT Prolongation Atlas (DIQTA). Potential application of DIQTA was shown by precision dosing in off-label use and therapeutic strategy optimization, as well as the facilitation of artificial intelligence (AI)-based modeling in predictive toxicity.

Keywords: Drug-induced prolongation of the QT interval; Torsades de pointes; Cardiotoxicity; Drug safety; Adverse drug reaction

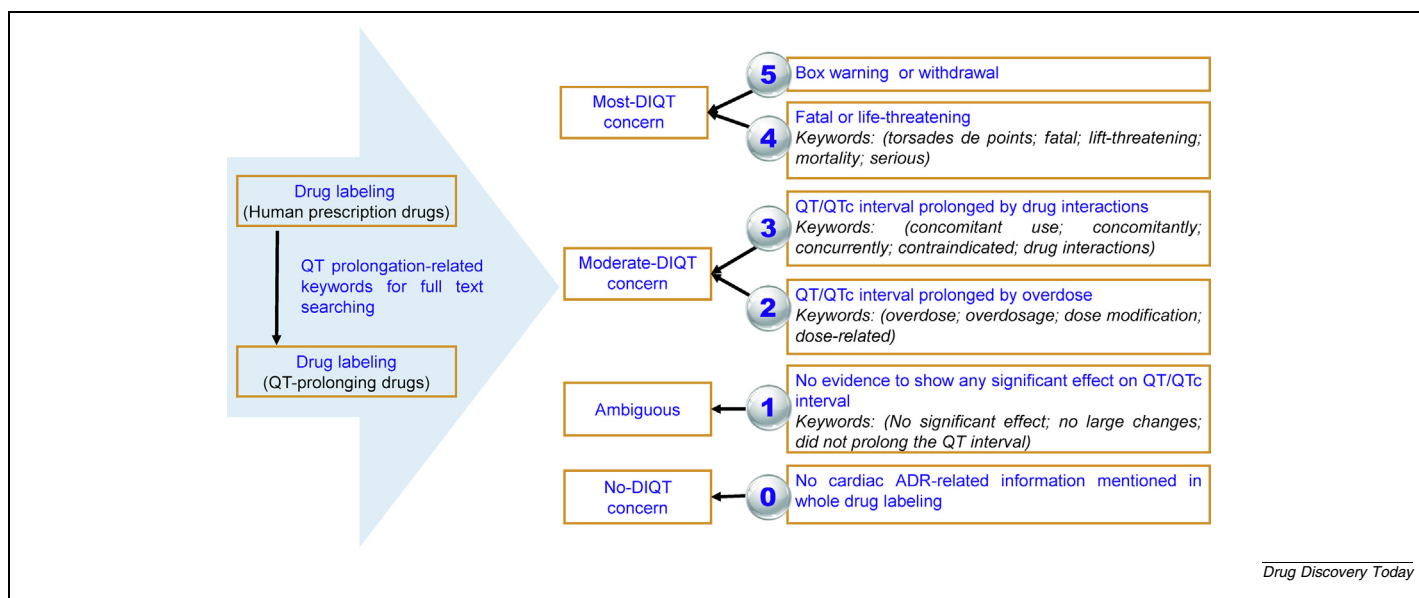
Introduction

With the emergence of new regimens and medications, cardiotoxicity becomes an important complication and can increase the risk of cardiac dysfunction. Better understanding of its molecular mechanisms and clinical relevance with therapies can be beneficial for preventing the occurrence of cardiovascular adverse events. In addition, the evidence of cardiotoxicity is also a considerable concern in new drug research and development for the pharmaceutical industry.

The QT interval on the electrocardiogram refers to the duration between the beginning of the QRS complex and the end of the T wave. Drug-induced prolongation of the QT interval is common in pharmaceutical treatments and can develop to a severe ventricular tachyarrhythmia named torsades de pointes (TdP), which can lead to cardiac sudden death. Although the incidence of TdP developed from QT interval prolongation is very low,¹ it has been a major reason for drug withdrawal from the market in the USA.² An increasing number of cases have been

reported to be associated with drug-induced QT interval prolongation, suggesting an urgent need for prevention and early detection in the drug discovery pipeline. Efforts have been made to assess and manage drugs with a potential risk of prolonging QT interval and inducing TdP.^{3–5} In clinical diagnosis, the increase in QT interval prolonged by a drug to >500 ms can be considered a strong risk signal exaggerating TdP.⁶ Genetic variations with torsadogenic risk have been identified in preclinical studies.⁷ A pharmacogenomics approach is proposed to discover novel biomarkers for predicting the drug response.^{8–9} Nevertheless, it is still difficult to predict the occurrence of TdP, especially in individualized drug therapy. To further improve the prediction of the occurrence of QT interval prolongation and TdP, a variety of data sources have been generated and applied. The CredibleMeds.org website provides a drug list in which the drugs are categorized into three subcategories based on their risk of TdP.¹⁰ The FDA maintains an electrocardiogram (ECG) warehouse for clinical data. In addition, the Comprehensive *In vitro*

* Corresponding author at: College of Chemistry, Sichuan University, Chengdu, Sichuan 610064, China. Wen, Z. (w_zhining@scu.edu.cn)

**FIGURE 1**

Schema for stratifying the risk of drug-induced prolongation of QT interval based on US Food and Drug Administration (FDA)-approved drug labeling. QT-prolonging drugs were extracted by full-text searching of drug labeling with QT prolongation-related keywords. Four concern levels (Most-DIQT concern, Moderate-DIQT concern, Ambiguous, and No-DIQT concern) were established according to the severity score of the drugs.

Proarrhythmia Assay (CiPA) initiative suggests integrating the response of multiple key cardiac ion channels to a drug to predict the global cardiotoxicity in stem cell-derived cardiomyocytes¹¹ and provides an open source data platform (<https://cipaproject.org/data-resources/>). The Health and Environmental Sciences Institute (HESI) Cardiovascular Pro-Arrhythmia Working Group (Pro-AWG) has generated a data set for assessing the predictive performance of nonclinical repolarization assays on predicting clinical QT study outcomes.^{12–13} Functional and pathological data are also applied to assess the cardiotoxicity of new candidate drugs.¹⁴ However, these data are scattered in the literature and pharmacovigilance databases. The lack of a centralized data resource for organizing, standardizing, and integrating information related to drug-induced prolongation of QT interval is a main hurdle to evaluating the potential TdP risk of drugs.

To fill this gap, we developed DIQTA to curate diverse types of information dispersed throughout the community. A risk stratification scheme was proposed for the marketed drugs to evaluate the potential risk of prolonging the QT interval based on FDA-approved drug labeling information. DIQTA not only provides a list of QT-prolonging drugs with stratified severity level, but also incorporates into the database: postmarketing surveillance data of QT-prolonging drugs extracted from the FDA FAERS database; drug property data, such as chemical structure; and host information, such as pharmacogenomics data. All the collated data can be easily accessed and downloaded from the web-based application (www.ADRAtlas.com/DIQT).

Risk stratification of drug-induced QT interval prolongation

Existing ways usually assess the risk of a QT-prolonging drug by counting the number of cases of QT interval prolongation in a

certain number of patients in response to a drug or by exploring the association of prolonged QT interval with the drug in a cohort study. We stratified the severity of QT-prolonging drugs based on FDA-approved drug labeling by considering the following advantages: (i) drug labeling includes essential scientific information for the safe and effective use of drugs, which is approved by FDA and regulated by law^{15–16}; (ii) the content of a drug label is determined after reviewing the data from clinical trials and postmarketing surveillance; and (iii) drug labels are a relatively accurate, stable, and update-to-date data source for annotating the risk of drug-induced QT interval prolongation. Following the successful development of the Liver Toxicity Knowledge Base (LTKB) led by the FDA^{17–18} and the Drug-Induced Rhabdomyolysis Atlas,¹⁹ we extracted QT-prolonging drugs from all human prescription drugs and established a scheme for risk stratification based on the drug-labeling information (Fig. 1). Details of labeling curation are described in the [Supplementary Materials](#) and Methods in the supplemental information online. In total, 428 drugs, including 413 marketed drugs with potential to cause QT interval prolongation and 15 drugs withdrawn from the market, were identified for further risk stratification.

The necessary information for the safe use of drugs is clearly declared in 18 labeling sections with specific headings, which are determined by the Code of Federal Regulations.¹⁵ The distribution of the information of QT interval prolongation across different labeling sections is shown in [Supplementary figure 1](#) in the supplemental information online. Of these drugs, 2.9% (12/413) will cause life-threatening arrhythmia and carry a warning in the 'Boxed warning' section. Over 20% drugs (98/413) described their risk of prolonging the QT interval in the 'Warnings and precautions' section, indicating the high severity of QT interval prolongation in their prescription. In addition, the

TABLE 1

Severity of DIQT based on descriptions in drug labeling.

Severity score	DIQT category	Keywords	Example description in drug labeling
5	Boxed warning or withdrawal	–	QT interval prolongation and serious arrhythmia (TdP) have occurred during treatment with methadone
4	Fatal or life-threatening	TdP, fatal, life-threatening, mortality, serious	Clinical studies indicate that trazodone hydrochloride might be arrhythmogenic in patients with pre-existing cardiac disease. Arrhythmias identified include isolated premature ventricular contractions, ventricular couplets, tachycardia with syncope, and TdP
3	QT/QTc prolonged by drug interaction	Concomitant use, concomitantly, concurrently, contraindicated, drug interactions	Overall analysis of ECG data in pediatric patients indicates that concomitant use of rocuronium bromide with general anesthetic agents can prolong QTc interval
2	QT/QTc prolonged by overdose	Overdose, overdosage, dose modification, dose-related	Symptoms of overdosage of primaquine phosphate include abdominal cramps, vomiting, burning epigastric distress, central nervous system and cardiovascular disturbances, including cardiac arrhythmia and QT interval prolongation, cyanosis, methemoglobinemia, moderate leukocytosis or leukopenia, and anemia in patients who are G6PD deficient
1	No significant effect on QT/QTc interval	No significant effect, no large changes, did not prolong QT interval	Administration of a single dose of acalabrutinib that is the fourfold maximum recommended single dose did not prolong QTc interval to any clinically relevant extent (i.e., ≥ 10 ms)

prolongation of QT interval is more likely to be caused by concomitant use and overdose, which are mentioned in drug–drug interaction sections [Drug interactions/contraindications, 23% (95/413) drugs] and dosage-related sections (Overdosage/dosage and administration, 22% (91/413) drugs], respectively.

We stratified the severity of drug-induced QT prolongation (DIQT) into five levels, scored from one to five according to the labeling content (Table 1). The highest severity score (severity score = 5) was assigned to drugs that were withdrawn from the market or carried a warning in the ‘Boxed warning’ in their drug label because they can induce fatal cardiac arrhythmia. High severity-level (severity score = 4) drugs referred to those that might cause fatal or life-threatening arrhythmia. The moderate severity level (severity score = 2 or 3) contained drugs that might prolong the QT interval by overdose or concomitant use. The lowest severity score (severity score = 1) was assigned to drugs without any significant effect on QT interval. We examined the distributions of the severity scores across seven labeling sections (Supplementary figure 2 in the supplemental information online). The percentage of the drugs with high severity score (4) was in descending order for the sections: ‘Warnings and precautions’ section (58%), dosage-related sections (30%), drug–drug interaction sections (25%) and the ‘Adverse reaction’ section (6%), which was consistent with the degree of seriousness of adverse drug reactions (ADRs) in sections defined by Title 21 of the Code of Federal Regulations 201.57.¹⁶

Subsequently, we proposed a determination of DIQT potential for a drug according to the severity score. The QT interval prolongation potential only reflected a matter of concern instead of a regulation decision and was determined by using the content of drug labeling information without involving individualized variations. In addition, we only used FDA-approved drug labeling to assign the severity score, which might be different when the drug labeling of other countries is used [e.g., European Union (EU) drug labeling]. The 428 QT-prolonging drugs were categorized into three concern levels: (i) Most-DIQT concern drugs might cause fatal or life-threatening arrhythmia (severity

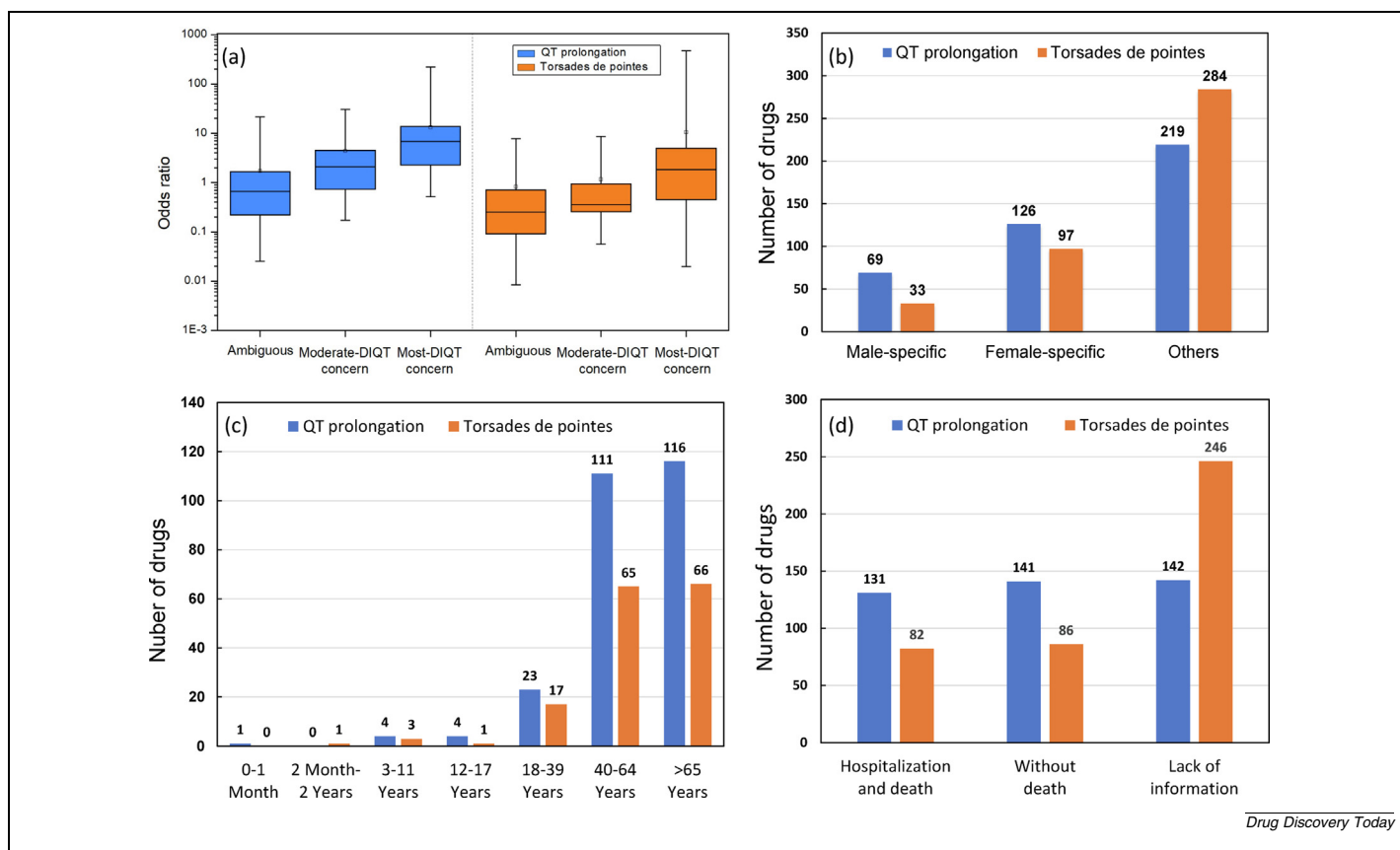
score = 4 or 5); (ii) Moderate-DIQT concern drugs might prolong the QT interval to some extent by overdose or concomitant use (severity score = 2 or 3); and (iii) the drugs that will not prolong the QT interval to any clinically relevant extent were placed in the ambiguous category (severity score = 1) (Fig. 1).

Furthermore, a No-DIQT concern drug list (severity score = 0) was curated, which included drugs without any cardiac ADR-related information. We first mapped the Preferred Terms (PTs) that represented standardized ADR information on their primary System Organ Classes (SOCs) in the *Medical Dictionary for Regulatory Activities* (MedDRA, www.meddra.org). Then, the PTs that belonged to ‘Cardiac disorders’ were extracted and used as keywords for searching drugs in the SIDER4.1 database.²⁰ Finally, we selected 100 drugs as No-DIQT drugs, which were not associated with any cardiac ADR-related PTs. In addition, we considered the distribution of QT-prolonging drugs in drug therapeutic categories and the drug physicochemical properties when identifying No-DIQT drugs. These drugs can be used as negative controls for constructing models for cardiotoxicity prediction.

Following the steps above, a benchmark DIQTA data set with 528 drugs was established, including 110 Most-DIQT concern drugs, 56 Moderate-DIQT concern drugs, 262 ambiguous drugs and 100 No-DIQT concern drugs (Table S1 in the supplemental information online).

Postmarketing surveillance data for drug-induced QT prolongation and TdP

Postmarketing surveillance data provide significant support for monitoring drug safety and managing idiosyncratic ADR. In this study, we extracted the case reports of the 413 QT-prolonging drugs from the FAERS database (<https://open.fda.gov/data/faers/>), which could be helpful for demonstrating the severity of these drugs and generating hypotheses for further verification in mechanistic studies. The overview of the distributions of odds ratios (ORs), gender difference, age distribution, and clinical outcomes of 413 QT-prolonging drugs are shown in Fig. 2 (details in

**FIGURE 2**

Postmarketing surveillance data of 413 QT-prolonging drugs for identifying population susceptibility to QT prolongation and torsades de pointes. **(a)** The distributions of odds ratios (ORs) in the three risk categories; **(b)** gender difference; **(c)** age distribution; **(d)** clinical outcomes.

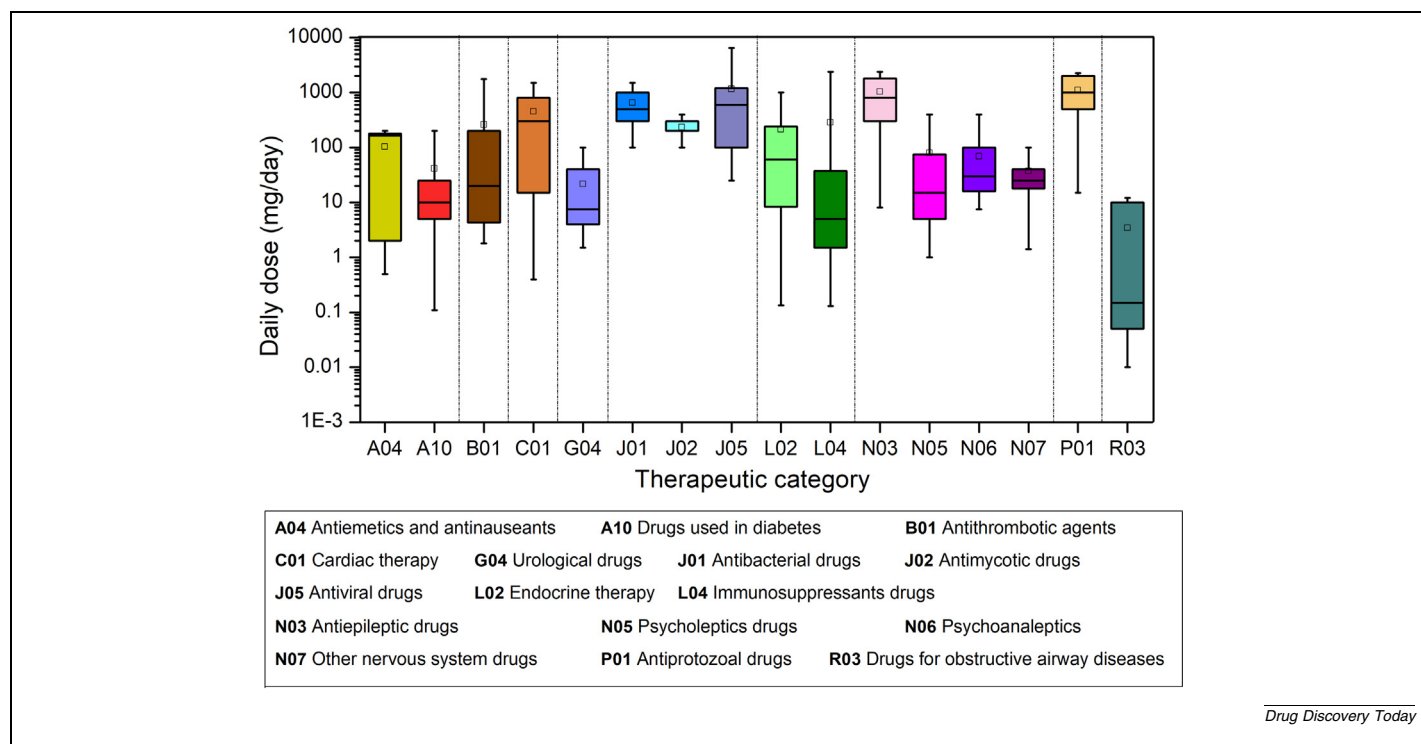
Table 2 in the supplemental information online). Given the limited data in the FAERS database, more emphasis on the distribution of patient data, especially racial/genetic differences, would also have a useful role in the investigation of drug-induced QT prolongation.

Fig. 2a shows the distributions of ORs in the three risk categories. Whether for QT prolongation or TdP, the average OR of Most-DIQT drugs was always the highest among the three categories. Several drugs in the ambiguous category, such as donepezil, exhibited relatively high ORs to induce QT prolongation and TdP, which highlighted the need for further safety evaluation for their DIQT potential. More QT-prolonging drugs were female specific (Fig. 2b), especially in terms of inducing TdP. In addition, 97 drugs were more likely to induce TdP in females, which was almost three times the number of drugs inducing TdP in males. Furthermore, patients in the oldest age group were more susceptible to developing QT interval prolongation and TdP (Fig. 2c). In addition, 28% (116/413) drugs would prolong the QT interval in patients over age 65. Although the ECG data of patients can be applied to intervene promptly with treatment in a clinical setting, postmarketing surveillance data showed that 31.7% (131/413) drugs would cause severe clinical outcomes after prolonging the QT interval (Fig. 2d).

Precision dosing in off-label use

Given that action potential prolongation can be caused by the inhibition of the *HERG* gene by a drug, high-throughput screening assays have been established to evaluate the potential effect of a new chemical entity on QT interval. In recent years, a few drugs have been found to cause TdP when following an FDA-approved (on-label) use. The standard dosing regimens for QT-prolonging drugs is suggested by the FDA, but it remains challenging to optimize dosing for patient subpopulations²¹ and reduce the incidence of clinical TdP caused by repurposing and non-FDA-approved (off-label) use of existing torsadogenic drugs.

For instance, chloroquine and hydroxychloroquine are approved for the treatment and prophylaxis of malaria with high recommended dosage and a need to avoid overdoses. At present, these two drugs are being used for the treatment of coronavirus disease 2019 (COVID-19) for which their associated risks and efficacies are under investigation.^{22–23} The urgent need for guidance for navigating the use of QT-prolonging drugs as a COVID-19 treatment has been raised to prevent QT prolongation and TdP, even though the risk of cardiac adverse reactions induced by chloroquine and hydroxychloroquine in patients with COVID-19 is currently unknown. Another antiprotozoal drug, quinine (quinine sulfate), might also induce cardiovascular adverse

**FIGURE 3**

Distributions of the drug daily dose (DDD) of 413 QT-prolonging drugs across 16 Anatomical Therapeutic Chemical (ATC) codes. The human daily dose was extracted from WHO ATC/DDD Index 2021. We used the second level of the ATC Classification System to represent the therapeutic categories of drug-induced QT prolongation (DIQT) drugs. The corresponding daily dose was separately extracted for a QT-prolonging drug used in multiple therapeutic categories.

events when off used -label for night-time leg cramps.²⁴ Another example is the off-label use of N03-antiepileptic drugs, which exhibit relatively higher daily doses. In pediatric neurology, it is still under discussion whether newer antiepileptic drugs (e.g., levetiracetam) should be used off-label for the treatment of neonatal seizures when safety information is lacking for children under 4 years of age.²⁵ Thus, determination of the initial dose for these drugs should be carefully conducted by neonatologists.

DIQTA collected the dose-related information of 413 QT-prolonging drugs from FDA-approved drug labeling. Of these, 18.4% (76/413) of drugs in DIQTA might cause QT prolongation if overdosed, as indicated in the drug labeling (Table S3 in the supplemental information online). In addition, the drug daily dose (DDD) information of 413 QT-prolonging drugs was extracted from ATC/DDD Index 2021 (www.whocc.no/atc-ddd-index/) and can be retrieved from DIQTA for further investigation. The dosage variance landscape of QT-prolonging drugs across different therapeutic categories is shown in Fig. 3 (details in Table S3 in the supplemental information online). These data could be helpful for optimizing dosing in off-label use. Although DIQTA only involves drugs administered by oral or parenteral routes, the route of administration is often crucial for pharmacological endpoints and should be fully considered in off-label use.

Therapeutic strategy optimization

Rapid development of cancer therapeutics dramatically improves the clinical outcomes and prognosis of patients with cancer.

Meanwhile, cardiotoxicity accompanying cancer treatment is an important concern when determining the appropriate therapeutic strategy. Severe cardiac arrhythmia might result from combination therapies because of interactions between two QT-prolonging drugs or the metabolism of both drugs through cytochrome P450 enzymes.^{26–27} For instance, as a second-generation multi-targeted tyrosine kinase inhibitor for the treatment of chronic myeloid leukemia, nilotinib can prolong the QT interval and cause sudden death if concomitantly used with other QT-prolonging drugs or strong CYP3A4 inhibitors. However, information about the QT prolongation induced by concomitant use is scattered and difficult to retrieve. Thus, it remains a considerable challenge for cardiologists and oncologists to ensure the efficacy of cancer therapies while reducing the risk of cardiotoxicity.

DIQTA collected all QT-prolonging drugs used in cancer therapies and showed that 29 of them should be avoided in terms of their concomitant use with other QT-prolonging drugs or CYP3A4 inhibitors (Table S4 in the supplemental information online). Further investigation of real-world data for the concomitant use of these drugs with other QT-prolonging drugs might indicate a need for therapeutic strategy optimization. We extracted the top-ten concomitantly used drugs from FAERS case reports for each of the 29 anticancer drugs mentioned above and found that several drugs, such as crizotinib, dasatinib, and glasdegib, are often concomitantly used with the QT-prolonging drug ondansetron. This might increase the risk of QT prolongation and induce TdP, and physicians should be aware of this

when determining therapeutic strategies.²⁸ The full list of concomitantly used drugs for each drug in DIQTA can be found in our web application (www.ADRAtlas.com/DIQTA). This information could be not only a support for clinical decision-making, but also an indication for designing *in vitro* toxicology assays to screen new chemical entities in preclinical studies. In addition, the consideration of disease states could be useful for optimizing the therapeutic strategies. Twelve of the 29 anticancer drugs might prolong the QT interval when being used for patients with specific disease states, such as congenital or acquired long QT syndrome, hypocalcemia, hypokalemia, hypomagnesemia, and severe hepatic impairment.

Artificial intelligence-based modeling for cardiotoxicity

The development of AI provides an excellent opportunity for the prediction of drug-induced prolongation of the QT interval. As an important branch of AI, machine learning (ML) methods exhibited a superior ability to predict the risk of drug-induced cardiotoxicity based on multiple data sources,²⁹ including chemical structural information, cellular-based data from *in vitro* bioassays, and ECG data from patients. DIQTA provides a well-stratified data set including 428 DIQT-concern drugs (involving 15 drugs withdrawn from the market) as positives and 100 No-DIQT concern drugs as negatives for modeling. For each QT-prolonging drug, the various types of information across preclinical studies, clinical trials, and postmarketing surveillance have been collated. All the data can be easily accessed and downloaded from a web-based application. This could facilitate the ADR research in drug development and improve drug attrition resulting from drug-induced QT prolongation. First, the diverse types of information provide rich data sources to use to construct ML models for the prediction and early detection of drug-induced QT prolongation. Second, the drug-labeling information contains cardiac ADRs other than QT interval prolongation. It should be possible to extend the mechanism of QT prolongation or TdP to predict other forms of cardiotoxicity by investigating the associations between QT-prolonging drugs and ADRs. Third, the data across different stages of drug development could facilitate the selection of QT-prolonging drugs for the design of new high-throughput screening assays in preclinical studies. Finally, it might be helpful to avoid cardiac ADRs when designing the prototype drug by comparing the information relating to QT-prolonging drugs at each stage.

Concluding remarks

A centralized data source with integrated risk factors of DIQT will be useful for further studying the underlying mechanisms, developing an effective prediction strategy, and supporting clinical decision-making. Based on FDA-approved drug labeling, DIQTA stratified the risk of 428 DIQT drugs and provides diverse types of information for preventing QT prolongation in a clinical setting as well as predicting the DIQT potential of new chemical entities in early drug development. The strategy for risk stratification is reproducible and the QT-prolonging drug list can be expanded as new FDA-approved drugs reach the market. To tackle the lack of negative samples in its modeling, DIQTA included 100 No-DIQT concern drugs as negative samples. However, to generate better predictive models, it would be useful to include more No-DIQT concern drugs. To the best of our knowledge, DIQTA is the first web-based application for providing comprehensive information about drug-induced QT interval prolongation and a one-stop solution to query/download the information for mechanism studies and predictive model development.

Caveats of our study should also be discussed. The description of safety information on QT prolongation is only moderately consistent between EU and US drug labeling.³⁰ Thus, the risk stratification of drugs might be different when using EU drug labeling. Although the QT-prolonging properties are described more explicitly in US drug labeling, ambiguous language is still used, especially for drugs without any significant effect on prolonging the QT interval. Therefore, several drugs in DIQTA are classified in the ambiguous category and more data are needed to evaluate their QT-prolonging potential. To prevent drug-induced TdP, more real-world data, such as electronic medical records data, should be incorporated because of a lack of incidence information in drug labeling. Our web-based application can be further improved based on users' suggestions. Thus, we hope that the curated DIQTA can enhance cardiac adverse reaction management and promote drug safety evaluation.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.drudis.2021.10.009>.

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